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B. Rushi Kumar · S. Ponnusamy ·
Debasis Giri · Bhavani Thuraisingham ·
Christopher W. Clifton ·
Barbara Carminati *Editors*

Mathematics and Computing

ICMC 2022, Vellore, India, January 6–8



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Contents

Computer Science

Color Multiscale Block-ZigZag LBP (CMB-ZZLBP): An Efficient and Discriminant Face Descriptor	3
Shekhar Karanwal	
Effect of Noise in the Quantum Network Implementation of Cop and Robber Game	15
Anjali Dhiman and S. Balakrishnan	
Study of Decoherence in Quantum Cournot Duopoly Game Using Modified EWL Scheme	27
A. V. S. Kameshwari and S. Balakrishnan	
A New Aggregation Operator for Single-Valued Triangular Neutrosophic Number in Decision-Making	37
G. Tamilarasi and S. Paulraj	
Redundancy of Codes with Graph Constraints	51
Ghurumuruhan Ganesan	
Tree Parity Machine-Based Symmetric Encryption: A Hybrid Approach	61
Ishak Meraouche, Sabyasachi Dutta, Haowen Tan, and Kouichi Sakurai	
Metadata Analysis of Web Images for Source Authentication in Online Social Media	75
Mohd Shaliyar and Khurram Mustafa	
A Computational Diffie–Hellman-Based Insider Secure Signcryption with Non-interactive Non-repudiation	89
Ngarenon Togde and Augustin P. Sarr	

kNN-SVM with Deep Features for COVID-19 Pneumonia Detection from Chest X-ray	103
Aman Bahuguna, Deepak Yadav, Apurbalal Senapati, and Baidya Nath Saha	
Quantum Simulation of Perfect State Transfer on Weighted Cubelike Graphs	117
Jaideep Mulherkar, Rishikant Rajdeepak, and Sunitha VadivelMurugan	
Quadratically Sound Proof-of-Sequential-Work	129
Souvik Sur and Dipanwita Roychowdhury	
On Some Properties of K-type Block Matrices in the Context of Complementarity Problem	143
A. Dutta and A. K. Das	
Families of Mordell Curves with Non-trivial Torsion and Rank of at Least Three	155
Renz Jimwel S. Mina and Jerico B. Bacani	
Applied Algebra and Analysis	
On the Genus of the Annihilator-Ideal Graph of Commutative Ring	165
Selvakumar Krishnan and Karthik Shunmugaiah	
The Radio k-chromatic Number for the Corona of Arbitrary Graph and K_1	175
P. K. Niranjana	
Some Parameters of Restricted Super Line Graphs	185
Latha Devi Puli and K. Manjula	
Edge Constrained Eulerian Extensions	195
Ghurumuruhan Ganesan	
Bounds of Some Energy-Like Invariants of Neighbourhood Corona of Graphs	205
Chinglensana Phanjoubam and Sainkumar Mn. Mawiong	
Linear Recurrent Fractal Interpolation Function for Data Set with Gaussian Noise	217
Mohit Kumar, Neelesh S. Upadhye, and A. K. B. Chand	
C^1-Rational Quadratic Trigonometric Spline Fractal Interpolation Functions	229
Vijay and A. K. B. Chand	
Cyclic Multivalued Iterated Function Systems	245
R. Pasupathi, A. K. B. Chand, and M. A. Navascués	

On Almost Statistical Convergence of Weight r	257
Ekrem Savaş	
Non-neighbor Topological Indices on Covid-19 Drugs with QSPR Analysis	263
W. Tamilarasi and B. J. Balamurugan	
Some Results on Differential Polynomials of Meromorphic Functions Sharing Certain Values	279
M. Tejuswini and N. Shilpa	
A Subclass of Pseudo-Type Meromorphic Bi-Univalent Functions of Complex Order Associated with Linear Operator	293
Asha Thomas, Thomas Rosy, and G. Murugusundaramoorthy	
Bi-Starlike Function of Complex Order Involving Double Zeta Functions in Shell Shaped Region	305
V. Malathi and K. Vijaya	
Fuzzy Rule-Based Expert System for Multi Assets Portfolio Optimization	319
Garima Bisht and Sanjay Kumar	
Stability Analysis of Additive Time-Varying T–S Fuzzy System Using Augmented Lyapunov Functional	335
Bhuvaneshwari Ganesan and Manivannan Annamalai	
Fractional Calculus and Integral Equations	
Solution of Fractional Differential Equations by Using Conformable Fractional Differential Transform Method with Adomian Polynomials	349
R. S. Teppawar, R. N. Ingle, and R. A. Muneshwar	
Generalized Results on Existence & Uniqueness with Wronskian and Abel Formula for α-Fractional Differential Equations	363
R. A. Muneshwar, K. L. Bondar, V. D. Mathpati, and Y. H. Shirole	
Method of Directly Defining the Inverse Mapping for Nonlinear Ordinary and Partial Fractional-Order Differential Equations	379
Dulashini Karunarathna and Mangalagama Dewasurendra	
Existence Results for Nonlocal Impulsive Fractional Neutral Functional Integro Differential Equations with Bounded Delay	391
M. Latha Maheswari and R. Nandhini	
An Application of Conformable Fractional Differential Transform Method for Smoking Epidemic Model	399
G. Tamil Preethi, N. Magesh, and N. B. Gatti	

Solvability of Infinite System of Volterra Integral Equations in the Tempered Spaces	413
Rahul and N. K. Mahato	
On Generalizations of Integral Inequalities and Its Applications	425
S. G. Latpate and S. V. Babar	
Mathematical Modelling and Fluid Dynamics	
Solving Multi-objective Chance Constraint Quadratic Fractional Programming Problem	441
Berhanu Belay and Adane Abebaw	
Higher Order Variational Symmetric Duality Over Cone Constraints	453
Sony Khatri and Ashish Kumar Prasad	
On Generalized Energy Inequality of the Damped Navier–Stokes Equations with Navier Slip Boundary Conditions	465
Subha Pal and Duranta Chutia	
The Diathermic Oils Over a Thin Liquid Film with MOS₂ Nano Particles: A Model with Analysis of Shape Factor Effects	479
S. Suneetha, K. Subbarayudu, and P. Bala Anki Reddy	
Wave Energy Dissipation by Multiple Permeable Barriers in Finite Depth Water	497
Biman Sarkar and Soumen De	
Thermal Stress Analysis of Inhomogeneous Infinite Solid to 2D Elasticity of Thermoelastic Problems	509
Abhijeet Adhe and Kirtiwant Ghadle	
Study of Non-Newtonian Models for 1D Blood Flow Through a Stenosed Carotid Artery	523
Mahesh Udupa and Sunanda Saha	
Two-Layered Flow of Ionized Gases Within a Channel of Parallel Permeable Plates Under an Applied Magnetic Field with the Hall Effect	541
M. Nagavalli, T. LingaRaju, and Peri K. Kameswaran	
Influence of Heat Transfer, Chemical Reaction and Variable Fluid Properties on Oscillatory MHD Couette Flow Through a Partially-Porous Channel	557
Sreedhara Rao Gunakala, Victor M. Job, and Jennilee Veronique	
Effect of Heat Transfer on Peristaltic Transport of Prandtl Fluid in an Inclined Porous Channel	573
Indira Ramarao, Priyanka N. Basavaraju, and Jagadeesha Seethappa	

A Multiscale Model of Stokes–Cahn–Hilliard Equations in a Porous Medium: Modeling, Analysis and Homogenization	591
Nitu Lakhmara and Hari Shankar Mahato	
Sensitivity and Directional Analysis of Two Mutually Competing Plant Population Under Allelopathy Using DDE	605
Dipesh and Pankaj Kumar	
Pore Scale Analysis and Homogenization of a Diffusion-Reaction-Dissolution-Precipitation Model	621
Nibedita Ghosh and Hari Shankar Mahato	
Mathematical Modeling and Computing to Study the Influence of Quarantine Levels and Common Mitigation Strategies on the Spread of COVID-19 on a Higher Education Campus	637
Raina Saha, Clarissa Benitez, Krista Cimbalista, Jolypich Pek, and Padmanabhan Seshaiyer	
Numerical Analysis	
Numerical Solution of the Fredholm Integral Equations of the First Kind by Using Multi-projection Methods	655
Subhashree Patel, Bijaya Laxmi Panigrahi, and Gnaneshwar Nelakanti	
Local Convergence of a Family of Kurchatov Like Methods for Nonlinear Equations	669
Abhimanyu Kumar and Soni Kumari	
An Effective Scheme for Solving a Class of Second-Order Two-Point Boundary Value Problems	681
Saurabh Tomar, Soniya Dhama, and Kuppalapalle Vajravelu	
An Analytic Solution for the Helmholtz-Duffing Oscillator by Modified Mickens' Extended Iteration Procedure	689
M. M. Ayub Hossain and B. M. Ikramul Haque	
Crank-Nicolson Finite Difference Scheme for Time-Space Fractional Diffusion Equation	701
Kalyanrao C. Takale and Veena V. Sangvikar (Kshirsagar)	
Gauss-Newton-Secant Method for the Solution of Non-linear Least-Square Problems Using ω-Condition	711
Naveen Chandra Bhagat, P. K. Parida, Chandresh Prasad, Sapan Kumar Nayak, Babita Mehta, and P. K. Sahoo	

The Diathermic Oils Over a Thin Liquid Film with MOS_2 Nano Particles: A Model with Analysis of Shape Factor Effects



S. Suneetha , K. Subbarayudu , and P. Bala Anki Reddy

Abstract A mathematical model is envisioned to depict and search out the report for different shapes of MOS_2 nanoparticles in a Casson nanofluid over an unsteady exponentially stretching sheet. The solid nanoparticles of Molybdenum disulphide are employed in different geometries such as bricks, cylinders, platelets, and blades in a porous medium. Also, Diathermic oil finds a remarkable application in mechanical engineering and industrial fields. By considering a non-uniform heat source/sink, it is possible to improve the rate of transferring of heat in diathermic oils, primarily Kerosene oil (KO) and Engine oil (EO). MATLAB's `bvp4c` function is used to compute the dimensionless forms of regulating flow expressions numerically. The role of relevant parameters on the fluid flow and heat transfer are debated by graphs and tables. It is significant that the heat transfer rate is more for blade-shaped MOS_2 nanoparticles when compared to other shapes.

Keywords and Phrases Thin liquid film · Shape effect · MOS_2 · Homogeneous—heterogeneous reactions · Porous media

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1 Literature Assessment

Heat transmission problem over a thin liquid film flow on an extending surface which is not steady has got vast applications in diverse fields like engineering, medical, and industrial themes such as wire varnishing, fiber coating, biophysics, films of tear in a human's eye, and also the condition of flow in a human's lungs and lubrication problems, are all unpredictable. Wang [1] was the first to discuss flow over a thin film on an extended sheet. Later, Andersson et al. [2, 3] adapted it for numerous physical natures of power-law fluid. However, Magyari and Keller [4] studied the effects of a stretching sheet on the flow field and boundary layers with exponentially rising velocity and temperature. As a result, many problems on an exponentially stretching sheet have been done by many investigators [5–8]. Nanofluid has nano-sized particles made of non-metallic or metallic materials and these fluids became a crucial item for researchers in many fields such as nano-drug delivery, microelectronics, nuclear power plants, heat exchanger, etc. The study of the mixed bag of fluid flow and electromagnetism is acknowledged as magneto-hydrodynamics (MHD), which is mainly used in the medical field, geophysics, designing MHD pumps, etc. Some motivating works on these nanofluids can be found in [9–11]. Molybdenum disulfide (MOS_2) is an inorganic compound made of layer by layer of molybdenum and sulphur atoms. Moreover, it is used in many mechanical applications due to its lubrication ability, slight frictional property, and robustness. Thermophysical properties depend on the size, shape, and volume fraction of nanoparticles and base fluid. Here four dissimilar nanoparticle shapes specifically cylinder, platelet, blade, and brick of MOS_2 are used. Hamilton and Crosser [12] observed an adequate amount of enhancement in the thermal conductivities with different nanoparticle shapes. Some other attempts on MOS_2 nanofluids with different shape effects are those made by [13–17]. In some chemically reacting processes like ignition, biochemical structures, explosion of fire-works, digestion of food, catalysis, and so forth homogeneous and heterogeneous reactions arise. Merkin [18] did the initial study on these reactions. Many scholars fed light on such type of reactions and are cited in Refs. [19–21].

Motivated by the above-cited literature, an effort has been made on a liquid thin film with EO and KO as the base fluids. MOS_2 nanoparticles of various shapes like cylinder, platelet, blade, and brick are appended in two separate individual source fluids. The model is resolved in MATLAB using the RK4S approach-bvp4c codes. The final results are displayed with graphs and tabular forms. This review extends the earlier studies and is also useful for different nanoparticle shape effects studies.

2 Formulation

In this work, a 2D porous nano liquid film of Magneto hydrodynamic nanofluid past an unsteady exponentially stretching sheet is considered. Here, MOS_2 is treated as a based nanoparticle. The stretching sheet switches on from a thin slot which is traced

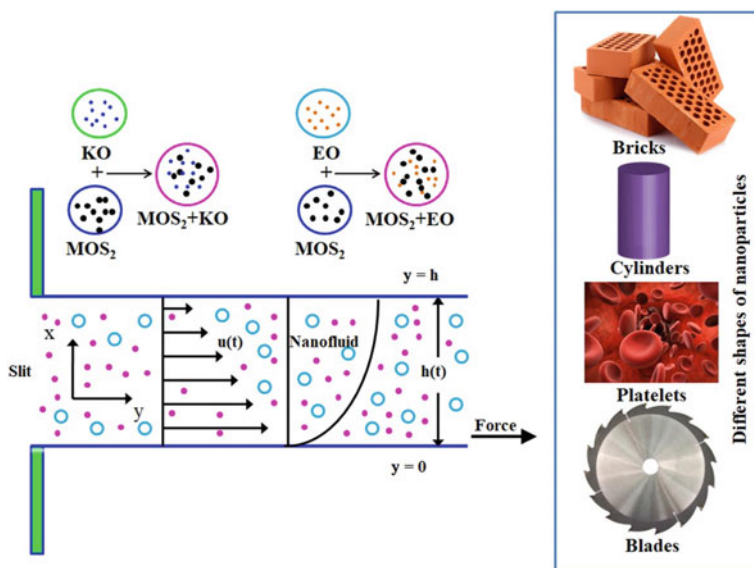


Fig. 1 Flow framework

at the initial point of a two-dimensional coordinates system (x, y) . The sheet stretches in the x -direction with velocity $U_w = \frac{cx}{1-\alpha t} e^{\frac{x}{l}}$ wherein c and α are not changeable and positive c denotes expanding rate and $\alpha t < 1$. The plane stretches erect to the y -axis. A magnetic field of strength $B(t) = \frac{B_0}{(1-\alpha t)^{\frac{1}{2}}}$ is applied perpendicularly to the sheet on the outside as revealed in Fig. 1. Reynolds number is very small so the induced magnetic field is insignificant. The sheet temperature is designed as $T_w = T_0 + T_{ref} \left(\frac{cx^2}{2\nu_f} \right) (1-\alpha t)^{-\frac{3}{2}} e^{\frac{x}{l}}$, where $\nu_f = \frac{\mu_f}{\rho_f}$ is the kinematic viscosity and f —the base fluid.

Chemical concentrations in the border flow for homogeneous and heterogeneous activities are assumed to be A^{**} and B^{**} .

$A^{**} + 2B^{**} \rightarrow 3B^{**}$, $rate = k_c ab^2$, $A^{**} \rightarrow B^{**}$, $rate = k_s a$, where k_c and k_s are the rate constants, a and b are the chemical species concentrations, and Isothermal reactions are supposed for both processes.

The flow with the above attention is in the form

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0 \quad (1)$$

$$\rho_{nf} \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) = \mu_{nf} \left(1 + \frac{1}{\beta} \right) \frac{\partial^2 u}{\partial y^2} - u \left(\sigma_{nf} B^2(t) + \frac{\mu_{nf}}{k^*} \right), \quad (2)$$

$$(\rho C_p)_{nf} \left(\frac{\partial T}{\partial t} + u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} \right) = k_{nf} \frac{\partial^2 T}{\partial y^2} + \frac{k_f U_w}{x \nu_f}$$

$$(A * (T_w - T_0)f'(\eta) + B * (T - T_0)) \quad (3)$$

$$\left(\frac{\partial a}{\partial t} + u \frac{\partial a}{\partial x} + v \frac{\partial a}{\partial y}\right) = D_A \left(\frac{\partial^2 a}{\partial x^2} + \frac{\partial^2 a}{\partial y^2}\right) + \frac{D_T}{T_\infty} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) - k_1 ab^2 \quad (4)$$

$$\left(\frac{\partial b}{\partial t} + u \frac{\partial b}{\partial x} + v \frac{\partial b}{\partial y}\right) = D_B \left(\frac{\partial^2 b}{\partial x^2} + \frac{\partial^2 b}{\partial y^2}\right) + \frac{D_T}{T_\infty} \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) + k_1 ab^2 \quad (5)$$

under the boundary conditions

$$u = U_w, v = V = 0, T = T_w, D_A \frac{\partial a}{\partial y} = -D_B \frac{\partial b}{\partial y} = ks \text{ at } y = 0 \quad (6)$$

$$\frac{\partial u}{\partial y} = \frac{\partial T}{\partial y} = 0, a = a_0, b = 0 \text{ at } y = H, v = \frac{\partial H}{\partial t} \text{ at } y = H(t) \quad (7)$$

here, $H(t)$ is the thickness of the fluid film.

Practically, the nanoparticle concentration of MOS_2 is little. With the help of Taylor's series, the nanofluid constants are

$$\begin{aligned} \frac{\mu_{nf}}{\mu_f} &= \frac{1}{(1 - 2.5\chi)}, \quad \frac{\rho_{nf}}{\rho_f} = 1 - \chi + \chi\Upsilon, \quad \frac{(\rho C_p)_{nf}}{(\rho C_p)_f} = 1 - \chi + \chi d, \\ \frac{k_{nf}}{k_f} &= 1 + \frac{3(k-1)\chi}{(k+2)}, \quad \frac{\sigma_{nf}}{\sigma_f} = \frac{3(\sigma-1)\chi}{(\sigma+2) - (\sigma-1)\chi} + 1 \\ \text{where } \quad i &= \frac{\rho_{sd}}{\rho_f}, \quad d = \frac{(\rho C_p)_{sd}}{(\rho C_p)_f}, \quad k = \frac{k_{sd}}{k_f}, \quad \sigma = \frac{(\sigma)_{sd}}{(\sigma)_f} \end{aligned} \quad (8)$$

In the present investigation, the effective thermal conductive k_{nf} can be estimated as [18]

$$\frac{k_{nf}}{k_f} = \frac{k_{sn} + (m-1)k_f + (m-1)(k_{sn} - k_f)\phi}{k_{sn} + (m-1)k_f - (k_{sn} - k_f)\phi}$$

Here, k_f and k_{sn} are the thermal conductivities of fluid and nanoparticles, respectively. The shape factor values are given in Table 1 (Table 2).

Table 1 Nanoparticle's shape factor (m)

Nanoparticles type	Shape factor (m)
Cylinders	4.9
Bricks	3.7
Blades	8.6
Platelets	5.7

Table 2 Shows the numerical values of the base fluids and nanoelements [21]

Physical properties	MOS ₂	EO KO	
C_p (J/kg K)	397.21	2048	2090
κ (W/m K)	904.4	0.1404	0.15
σ (s/m)	2.09×10^{-4}	55×10^{-6}	21×10^{-6}
ρ (kg/m ³)	5060	863	783

Unveiling the dimensionless variables as

$$\eta = y e^{\frac{x}{2l}} \left(\frac{c}{v_f(1-\alpha t)} \right)^{\frac{1}{2}}, \quad \psi = \beta_1 \left(\frac{c}{1-\alpha t} \right)^{\frac{1}{2}} f(\eta) e^{\frac{x}{2l}}, \quad \phi = \frac{a}{a_0}, \quad h = \frac{b}{a_0}$$

$$u = \beta_1 \frac{c}{1-\alpha t} x f'(\eta) e^{\frac{x}{l}}, \quad v = -\beta_1 \left(\frac{c v_f}{1-\alpha t} \right)^{\frac{1}{2}} f(\eta) e^{\frac{x}{2l}}, \quad T = T_0$$

$$+ T_{ref} \left(\frac{c x^2}{2 v_f} \right) (1-\alpha t)^{-\frac{3}{2}} \theta(\eta) e^{\frac{x}{2l}}, \quad (9)$$

Adopting Eqs. (8, 9) in Eqs. (1-5), we have

$$\phi_1 f''' \left(1 + \frac{1}{\beta} \right) + \phi_2 (\beta_1)^2 \left[f f'' - (f')^2 - S \left(f' + \frac{\eta}{2} f'' \right) \right] - \phi_3 M f' - D f' = 0 \quad (10)$$

$$\phi_4 \theta'' - \phi_5 \text{Pr}(\beta_1)^2 \left[2 f' \theta - f \theta' + \frac{3}{2} S \theta + \frac{1}{2} S \eta \theta' \right] + A * f' + B * \theta = 0 \quad (11)$$

$$\frac{1}{Sc} \left(\phi'' + \frac{1}{N_{AT}} (\theta + \theta'') \right) + f \phi' - S \phi' \frac{\eta}{2} - K \phi h^2 = 0 \quad (12)$$

$$\frac{\delta}{Sc} \left(h'' + \frac{1}{N_{AT}} (\theta + \theta'') \right) + f h' - S h' \frac{\eta}{2} + K \phi h^2 = 0 \quad (13)$$

$$\text{at } \eta = 0 \rightarrow f(0) = 0, \quad f'(0) = 1, \quad \theta(0) = 1, \quad \phi'(0) = K s \phi(0), \quad \delta h'(0) = -K s \phi(0) \quad (14)$$

$$\text{at } \eta = 1 \rightarrow f(1) = \frac{\beta_1 S}{2}, \quad f''(1) = 0, \quad \theta'(1) = 0, \quad \phi(1) = 1, \quad h(1) = 0 \quad (15)$$

where

$$\phi_1 = \frac{(\mu)_{nf}}{(\mu)_f}, \quad \phi_2 = \frac{\rho_{nf}}{\rho_f}, \quad \phi_3 = \frac{\sigma_{nf}}{\sigma_f}, \quad \phi_4 = \frac{k_{nf}}{k_f}, \quad \phi_5 = \frac{(\rho C_p)_{nf}}{(\rho C_p)_f},$$

$$\begin{aligned}
 K &= a_0^2 k_1 \frac{(1 - \alpha t)}{c}, \quad Ks = \frac{k_s}{D_A a} \left(\frac{c}{v_f(1 - \alpha t)} \right)^{-\frac{1}{2}}, \\
 S &= \frac{\alpha}{c}, \quad M = \frac{\sigma_f B_0^2}{c \rho_f}, \quad D = \frac{v}{\rho_f k^*} \frac{(1 - \alpha t)}{c}, \quad \text{Pr} = \frac{(\mu C_p)_f}{k_f}, \quad \delta = \frac{D_B}{D_A}, \\
 N_{AT} &= \frac{D_A a_0}{\frac{D_T}{T_\infty} (T_w - T_\infty)},
 \end{aligned} \tag{16}$$

Thus, $\delta = 1$ for D_A and D_B is equal $\phi(\eta) + h(\eta) = 1$,

Equations (12) and (13) under this assumption decomposed to

$$\frac{1}{Sc} \left(\phi'' + \frac{1}{N_{AT}} (\theta + \theta'') \right) + f\phi' - S\phi' \frac{\eta}{2} - K\phi(1 - \phi)^2 = 0 \tag{17}$$

related boundary conditions: $\eta = 0 \rightarrow \phi'(0) = Ks\phi(0)$, $\eta = 1 \rightarrow \phi(1) = 1$

The quantities of the engineering curiosity drag force on the surface and Nusselt number at local are given as

$$C_f = \frac{\tau_w}{\rho_f U_w^2}, \quad Nu = \frac{q_w x}{k_f (T_w - T_0)} \tag{18}$$

where τ_w (wall skin friction), and q_w (wall heat flux) are given as

$$\tau_w = \mu_{nf} \left(1 + \frac{1}{\beta} \right) \left(\frac{\partial u}{\partial y} \right)_{y=0}, \quad q_w = -k_{nf} \left(\frac{\partial T}{\partial y} \right)_{y=0} \tag{19}$$

In view of Eqs. (9) and (19) in Eqs. (18), we acquire

$$C_f = (\text{Re}_x)^{-\frac{1}{2}} \left(1 + \frac{1}{\beta} \right) \frac{1}{(1 - 2.5\chi)} f''(0), \quad Nu_x = -(\text{Re}_x)^{\frac{1}{2}} \frac{-1}{\beta_1} \frac{k_{nf}}{k_f} \phi'(0) \tag{20}$$

where $\text{Re}_x = \frac{U_w x}{\nu_f}$ signifies the local Reynolds number.

3 Plan of Solution

By transforming partial differential expressions into ordinary differential equations by using suitable transformations, which results highly nonlinear Eqs. (10), (11) and (17) which cannot be solved analytically. Therefore, we employ the famous shooting technique with RKF method. The set of coupled nonlinear ODEs is renovated into the system of differential equations of the first order as follows:

$$f = P_1, \quad f' = P_2, \quad f'' = P_3, \quad f''' = P'_3, \quad \theta = P_4, \quad \theta' = P_5, \quad \theta'' = P'_5, \quad \phi = P_6, \quad \phi'$$

$$\begin{aligned}
&= P_7, \phi'' = P'_7, \\
P'_3 &= \left(\frac{1}{\phi_1 \left(1 + \frac{1}{\beta} \right)} \right) (-\phi_2(\beta_1)^2 P_1 P_3 - P_2^2 - S(P_2 + 0.5\eta P_3)) + \phi_3 M P_2 + D P_2 \\
P'_5 &= \frac{1}{\phi_4} \left[\phi_5 \text{Pr}(\beta_1)^2 \left(2P_2 P_4 - P_1 P_5 + \frac{3}{2} S P_4 + \frac{1}{2} S \eta P_5 \right) + A * P_2 + B * P_4 \right] \\
P'_7 &= -\frac{1}{N_{AT}} (P_4 P'_5) + Sc(-P_1 P_7 + 0.5 S \eta P_7 + K P_6(1 - P_6)^2)
\end{aligned}$$

Associated boundary conditions are

$$\begin{aligned}
P_1(0) &= 0, P_2(0) = 1, P_4(0) = 1, P_7(0) = Ks P_6(0), P_1(1) = \frac{\beta_1 S}{2}, P_3(1) = 0, \\
P_5(1) &= 0, P_6(1) = 1
\end{aligned}$$

For solving boundary layer flow problems, this technique is very much useful. Assume two guesses $f''(0)$ and $-\theta'(0)$ to get an approximate solution. For all cases, the step size is 0.001, and the convergence criterion is 10^{-6} .

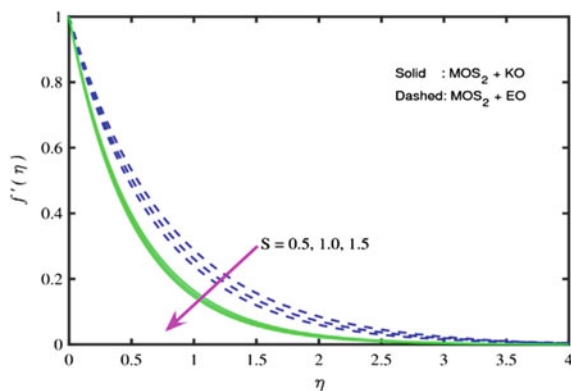
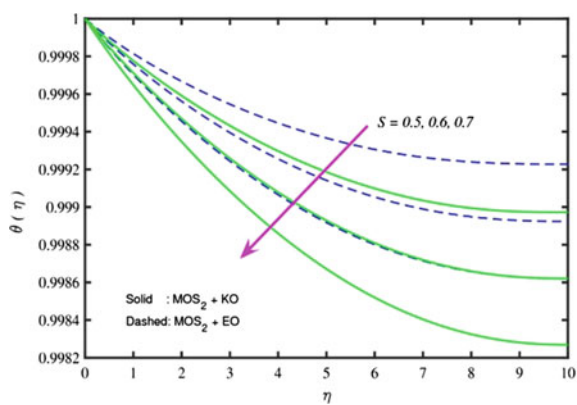
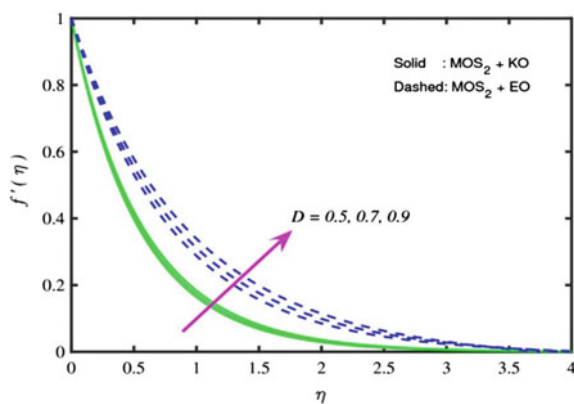
4 Discussion

This portion designates momentous features of tangled flow parameters on velocity, temperature and nano particle concentration, the rate of heat transfer, surface drag force, and local convective transmission coefficient. Computer codes have been propelled for these numerical results and are incited with graphs and tables assuming the default values for all the physical parameters $Sc = 0.5$, $K = 0.5$, $\beta_1 = 1.0$, $q = 0.5$,

$\phi_1 = 1.0$, $\phi_2 = 1.0$, $S = 0.5$, $\theta_w = 1.5$, $M = 0.5$, $\text{Pr} = 0.72$, $A* = 0.05$, $B* = 0.05$, $N_{AT} = 1.0$, and $Ks = 10.0$ are fixed. At this juncture, two cases: MOS₂ + KO and MOS₂ + EO are examined.

The upshot of the unsteadiness parameter S on flow and thermal is displayed in Figs. 2 and 3. It is reported that the upgrading values of S degrade the velocity and temperature profiles for two cases. It should also be noticed that MOS₂ + KO leads to better outcomes for the boundary layers of thermal and momentum than for the concentration. This is due to the influence of buoyancy on the fluid flow shrinks for escalating values of S . This results in a decrement in all three boundary layers.

Figure 4 fed light on the velocity variations for different porosity D on thin nanoliquid film. It is observed that velocity hikes as D hikes. The porosity should be increased slightly as the thickness of the film is very little, that is $D \rightarrow \infty$ corresponds to the case in which there does not exist any porous medium. The growing values of D stand for the massive opening of the permeable gap, resulting in the retardation of the flow with elevated velocity.

Fig. 2 $f'(\eta)$ for S **Fig. 3** $\theta(\eta)$ for S **Fig. 4** $f'(\eta)$ for D 

Concentration variations are displayed in Figs. 5 and 6 for changing the scale of the reactions K and Ks . As K and Ks values mounted a decrement in concentration is inferred from the figures. When the strength of homogeneous reactions is increased, the consumption of chemical reactants improves as well, resulting in a large chemical reaction and, as a result, a smaller concentration distribution. For various base fluids, the thickness of the solutal boundary layer decreases. Because the heterogeneous reaction parameter Ks has an opposite relation with mass diffusivity, the concentration falls.

Figure 7 displays the outcome of Sc on concentration. Sc refers to simultaneous momentum and mass diffusion convection in a fluid flow. Sc is the proportion of the rates of viscous diffusion to molecular diffusion. Low concentration for high Sc is noted. This is due to the fact that strong diffusion species have a greater retarding impact on the concentration.

Fig. 5 $\phi(\eta)$ for K

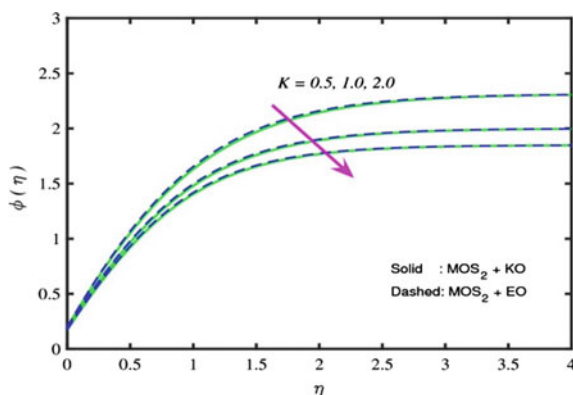


Fig. 6 $\phi(\eta)$ for Ks

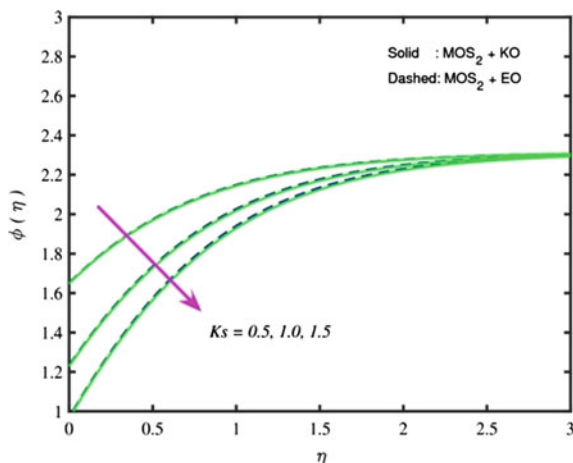


Fig. 7 $\phi(\eta)$ for Sc

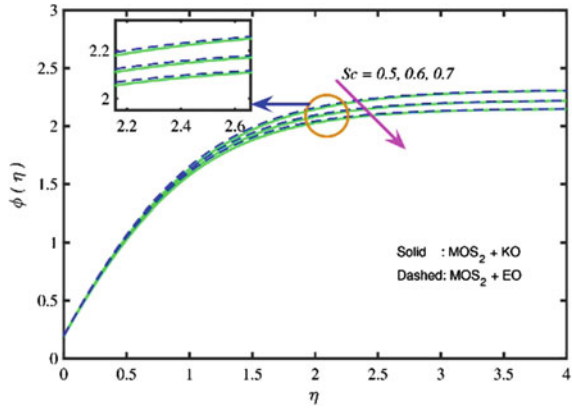


Figure 8 illustrates how the temperature distribution is affected by the space-dependent heat source/sink parameter A^* . The boundary layer generates energy if $A^* > 0$ (heat source), which boosts the liquid's thermal nature. The border layer absorbs energy for $A^* < 0$ (absorption), consequences a drop in temperature. In Fig. 9, the effect of B^* on temperature is shown. As energy is discharged into the fluid, a hike in fluid temperature is noticed when $B^* > 0$, and a negativity is noted, i.e., a declivity in heat as the energy is taken up by the fluid when $B^* < 0$.

Figure 10 emphasizes the variation in Surface drag force for different D . It brings out that escalating D the drag force is added for $MOS_2 + KO$ and $MOS_2 + EO$. The consequence of S on drag force is shown in Fig. 11. It is uncovered that rising S raises the drag force for both cases.

Figure 12 exemplifies the consequence of heat flux on A^* . It is viewed from the figure that a rise in A^* has a plunge in the heat flux. It is detected from Fig. 13 that the heat flux reduces with B^* . The increment for $MOS_2 + EO$ is further with $MOS_2 + KO$.

Fig. 8 $\theta(\eta)$ for A^*

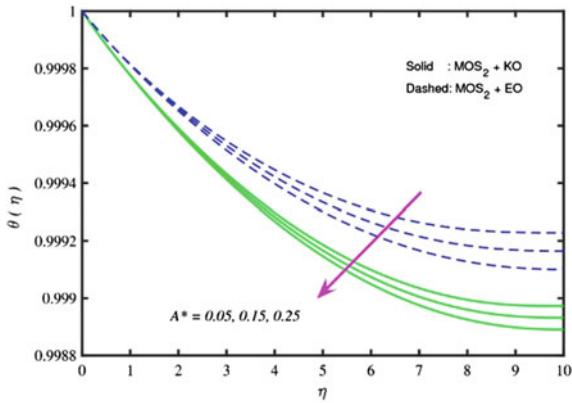
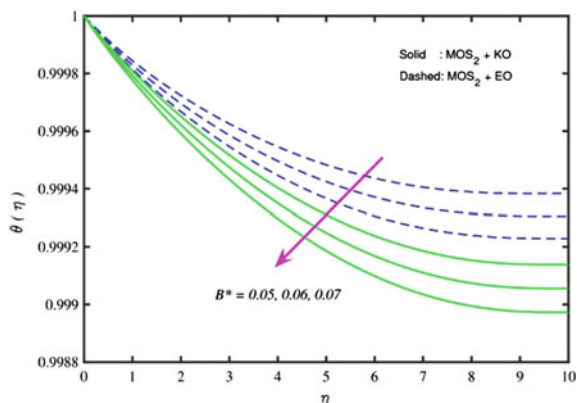
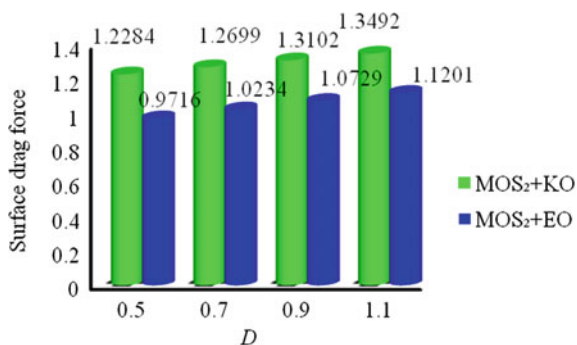
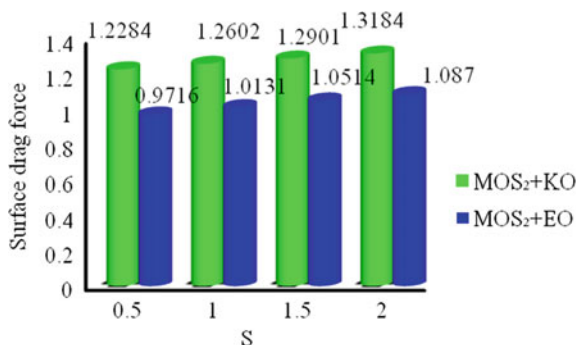
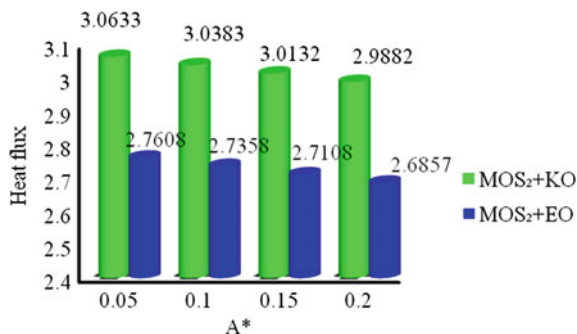
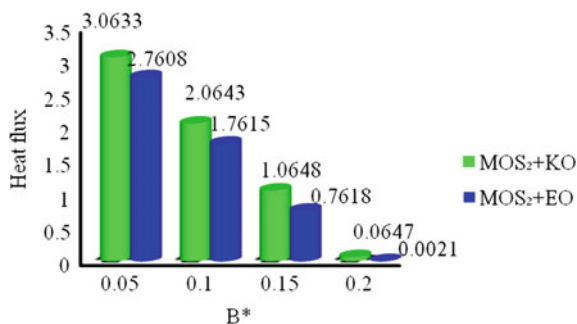


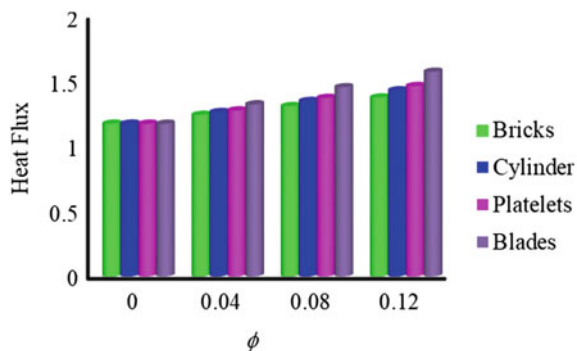
Fig. 9 $\theta(\eta)$ for B^* **Fig. 10** $f''(\eta)$ for D **Fig. 11** $f''(\eta)$ for S 

For different shapes of MOS₂ nanoparticles, the rate of heat transfer and volume fraction in kerosene oil is debated in Fig. 14. Here, the heat transfer rate amplifies with ϕ . The heat transfer rate of the nanofluid with particles of blade-shapes is bigger than brick, cylinder, and platelet shapes for the same volume fraction. The heat transfer rate with particles of blade shapes is 12.70, 23.93, and 34.07% greater

Fig. 12 $-\theta'(\eta)$ for A^* **Fig. 13** $-\theta'(\eta)$ for B^* 

than brick, cylinder, and platelet shapes in kerosene oil-based nanofluids. MOS₂ is more conductive when compared to other metals because of its special arrangement with diamagnetic nature. However, the heat transfer rate is low.

Figure 15 is displayed to report the increment percentage of unlike shapes of MOS₂ nanoparticles with Engine oil as a base fluid. The figure shows that the particles of blade shapes have the highest variation succeeded by brick, cylinder, and platelet shapes. Also, spotted that the thermal transfer rate of EO has been enhanced by 33.04% with blade-shaped particles. Also noted that 17.05%, 21.52%, and 24.28%

Fig. 14 $-\theta'(\eta)$ for ϕ with different shapes on MOS₂ + KO

increment of the thermal transfer rate of EO with brick, cylinder, and platelet-shaped nanoparticles, respectively. Table 3 shows a deviancy of surface drag force for various denominations of M , D , S , and β_1 . The drag force hastens with M , D , S and decelerates along β_1 . The increment of the rate of heat transfer with dissimilar shaped nanoparticles in EO and KO with divergent volume fractions are displayed in Tables 4 and 5. The blade-shaped particles show the utmost deviation caught on by brick, cylinder, and platelet-shaped particles in both base fluids.

Fig. 15 $-\theta'(\eta)$ for ϕ with different shapes on MOS₂ + EO

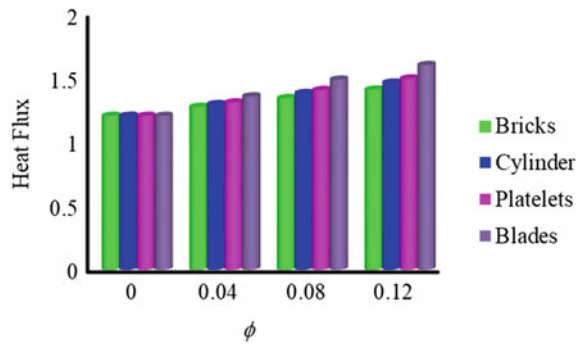


Table 3 The values of surface drag force for various parameters

				$f''(0)$	
M	D	S	β_1	MOS ₂ + KO	MOS ₂ + EO
0.5	0.5	0.5	0.5	1.2284	0.9716
1.0	0.5	0.5	0.5	1.5398	1.1278
1.5	0.5	0.5	0.5	1.7988	1.2653
2.0	0.5	0.5	0.5	2.0253	1.3894
0.5	0.7	0.5	0.5	1.2699	1.0234
0.5	0.9	0.5	0.5	1.3102	1.0729
0.5	1.1	0.5	0.5	1.3492	1.1201
0.5	0.5	1.0	0.5	1.2602	1.0131
0.5	0.5	1.5	0.5	1.2901	1.0514
0.5	0.5	2.0	0.5	1.3814	1.0870
0.5	0.5	0.5	1.0	0.7967	0.6901
0.5	0.5	0.5	1.5	0.6746	0.6106
0.5	0.5	0.5	2.0	0.6150	0.5692

Table 4 ϕ for $-\theta'(0)$ and percent enhancement with bricks, cylinders, platelets, and blades on $\text{MOS}_2 + \text{EO}$

ϕ	$-\theta'(0)$				%			
	Bricks	Cylinders	Platelets	Blades	Bricks	Cylinders	Platelets	Blades
0	1.2061	1.2061	1.2061	1.2061	—	—	—	—
0.04	1.2766	1.2978	1.3116	1.3589	5.84529	7.60302	8.7472	12.6689
0.08	1.3451	1.3841	1.4087	1.4900	11.5248	14.7583	16.7979	23.5387
0.12	1.4118	1.4657	1.4989	1.6046	17.055	21.5239	24.2766	33.0404

Table 5 ϕ for $-\theta'(0)$ and percent enhancement with bricks, cylinders, platelets, and blades on $\text{MOS}_2 + \text{KO}$

ϕ	$-\theta'(0)$				%			
	Bricks	Cylinders	Platelets	Blades	Bricks	Cylinders	Platelets	Blades
0	1.1752	1.1752	1.1752	1.1752	—	—	—	—
0.04	1.2434	1.2642	1.2777	1.3244	5.80327	7.57318	8.72192	12.6957
0.08	1.3106	1.3493	1.3740	1.4564	11.5214	14.81450	16.91630	23.9278
0.12	1.3771	1.4315	1.4655	1.5756	17.1801	21.80910	24.70220	34.0708

5 Conclusions

The central concluded points are as follows:

- The MOS_2 nanoparticle is more active in EO than KO.
- The non-spherical elements (Platelet and Cylinder) within EO–KO-based fluids conclude better viscosity owing to their configuration and will intensely increase the heat transport capacity.
- The Surface drag force is high for high Porosity parameter.
- Heat transfer rates of EO having blade-shaped particles are 12.67, 23.54, and 33.04%; superior to the regular fluid with volume fraction $\phi = 0.04, 0.08$ and 0.12 , respectively.
- The heat transfer rate of EO-based nanofluid suspended blade-shaped MOS_2 nanoparticles is 33.04, 17.05, 21.52, and 24.28%; bigger than brick, cylinder, and platelet-shaped nanoparticles.

Graphical Trends

(See Figs. 2 to 15. Table 2 to 5).

Nomenclature

e_{ij}	(I, j)th element of the deformation rate
B_0	Applied magnetic flux
$f(\eta)$	Dimensionless velocity
EO	Engine oil
q_w	Heat flux from the surface
q	Dimensionless parameter
K_s	Heterogeneous reaction strength
K	Homogeneous reaction strength
U_w	Stretching velocity in X direction (m s^{-1})
V	Stretching velocity in Y direction (m s^{-1})
$H(t)$	Film size (m)
T_w	Temperature of the fluid near the wall (K)
T_0	Initial temperature of the fluid (K)
KO	Kerosene oil
C_{f_x}	Local skin friction in dimensionless form along x-direction
Nu_x	Local Nusselt number
Re_x	Local Reynolds number
M	Magnetic field parameter
Pr	Prandtl number
p	Pressure ($\text{kg m}^{-1}\text{s}^{-2}$)
D	Porosity parameter
T_{ref}	Referred temperature of the fluid (K)
T	Temperature (K)
C_p	Specific heat at constant pressure ($\text{J kg}^{-1}\text{K}^{-1}$)
T_s	Temperature of the fluid over surface (K)
t	Time (s)
c	Stretching rate (s^{-1})
k	Thermal conductivity ($\text{m}^2 \text{s}^{-1}$)
S	Unsteadiness parameter
Sc	Schmidt number
u	Velocity component along x-axis (m s^{-1})
v	Velocity component along y-axis (m s^{-1})
x	x – Coordinate (m)
y	y – Coordinate (m)

Greek Symbols

α	Constant (s^{-1})
π_c	Critical value of the product of the deformation rate by itself
ρ	Density (kg m^{-3})

ψ	Physical stream function ($\text{m}^2 \text{s}^{-1}$)
δ	Ratio of diffusion coefficient
β_1	Dimensionless fluid thickness parameter
ν	Kinematic viscosity ($\text{m}^2 \text{s}^{-1}$)
$\phi_i (i = 1 - 5)$	-Nanofluids constants
τ_w	Surface shear stress ($\text{kg m}^{-1} \text{s}^{-2}$)
η	Similarity variable
σ	Electrical conductivity
$\theta(\eta)$	Dimensionless temperature
ϕ	Nano particles volume fraction of MOS_2

Subscripts

f	Base fluid
∞	Fluid properties at ambient flow
nf	Nanofluid
s	Surface

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